Heinrich Weber, *Lehrbuch der Algebra*, Volume I
Translation Batch 08: Inverses, Transforms, and Permutation Substitutions

English translation and mathematical normalization

Translator's note

This batch continues the translation of Weber's treatment of groups of linear substitutions. I have kept Weber's order of argument and his matrix notation, while normalizing a few terms: *Substitution* is retained where Weber means a linear change of variables; *composite*, *inverse*, *transpose*, and *transformed substitution* are used in their modern senses. The formulas are kept close to the German source.

Inverse substitutions

Composition with the identity substitution J changes no substitution. Thus, in the composition law, J is regarded as the unit element, and may be omitted whenever it occurs as a factor composed with other substitutions.

We now have the following theorem.

Theorem. For every substitution A there exists a unique inverse substitution A^{-1} satisfying

$$AA^{-1} = A^{-1}A = J. \tag{1}$$

This follows from the basic formulas of determinant theory. If the entries of A^{-1} are denoted by $\alpha_h^{(k)}$, then they are determined by the linear equations obtained from the requirement that the product with A be the identity. Since the determinant of A is assumed non-zero, these equations have a unique solution. The inverse substitution is therefore uniquely determined.

For inverses of composites one obtains, from

$$ABB^{-1}A^{-1} = J,$$

the formula

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{2}$$

If A, B, C are three substitutions, then from either of the equations

$$AB = AC, \qquad BA = CA,$$

composition with A^{-1} gives $B = C$. Thus the characteristic properties required for a group are satisfied under composition. The totality of all substitutions of a fixed dimension is therefore an infinite group. Moreover, if one selects from this totality any set closed under composition, then that set itself forms a group, finite or infinite according to the case.

Transformed substitutions

Let L be a fixed substitution. From any substitution A of the same dimension one obtains another substitution

$$A' = L^{-1}AL, \tag{3}$$

which is called the transform of A by L.

Putting

$$(x) = L(x'), \qquad (y) = L(y'), \tag{4}$$

one obtains, from $(y) = A(x)$,

$$(y') = A'(x'). \tag{5}$$

Consequently the passage to the transformed substitution is equivalent to applying the same change of variables, namely L^{-1}, to both systems of variables.

If

$$A' = L^{-1}AL, \qquad B' = L^{-1}BL,$$

then the laws of composition give

$$A'B' = L^{-1}ABL.$$

Hence:

Theorem. As A runs through the substitutions of a group, and L is kept fixed, the transformed substitutions $L^{-1}AL$ run through the substitutions of an isomorphic group.

Transposed substitutions

Inverse substitutions must be distinguished from transposed substitutions. From a substitution A one obtains a definite new substitution, denoted here by A_t, by turning rows into columns and columns into rows; equivalently, in Weber's notation, one interchanges the upper and lower indices of the coefficients of A.

If in the coefficient formula for the product $C = BA$ one interchanges upper and lower indices and simultaneously interchanges the letters a and b, one obtains the coefficients of the transpose of the product. Thus, when the factors in the product are transposed, their order is reversed:

$$(BA)_t = A_t B_t. \tag{6}$$

The operation of transposition is therefore formally similar to taking inverses, in that an order reversal occurs. Combining both operations gives

$$(A^{-1})_t = (A_t)^{-1}, \tag{7}$$

and, where no ambiguity is possible, this may be written simply as A_t^{-1}.

Linear fractional substitutions in one variable

We now apply the preceding general results to some special cases, beginning with substitutions of one variable. These substitutions are the same as the linear fractional functions

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \tag{8}$$

for which Weber introduces the abbreviation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{9}$$

Their composition is governed by

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha\alpha' + \beta\gamma' & \alpha\beta' + \beta\delta' \\ \gamma\alpha' + \delta\gamma' & \gamma\beta' + \delta\delta' \end{pmatrix}. \tag{10}$$

The coefficients $\alpha, \beta, \gamma, \delta$ are arbitrary numbers subject only to the condition that

$$D = \alpha\delta - \beta\gamma \neq 0.$$

The identity substitution is

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{11}$$

and the inverse of (9) is

$$\begin{pmatrix} \dfrac{\delta}{D} & \dfrac{-\beta}{D} \\ \dfrac{-\gamma}{D} & \dfrac{\alpha}{D} \end{pmatrix}. \tag{12}$$

The substitution transposed to (9) is obtained by interchanging the diagonal entries and changing the signs of the off-diagonal entries:

$$\begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}. \tag{13}$$

Equivalently, this is obtained by replacing

$$\alpha, \beta, \gamma, \delta \quad \text{by} \quad \delta, -\gamma, -\beta, \alpha.$$

One verifies immediately that

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & D \end{pmatrix}. \tag{14}$$

This both confirms the preceding rule for transposes and shows that every such substitution commutes with its transposed substitution.

Permutation groups as linear substitutions

Let $x_1, x_2, \dots, x_n$ be a system of n variables, and let

$$\alpha_1, \alpha_2, \dots, \alpha_n$$

be some ordering of the numbers $1, 2, \dots, n$. The equations

$$x_1 = x'_{\alpha_1}, \quad x_2 = x'_{\alpha_2}, \quad \dots, \quad x_n = x'_{\alpha_n} \tag{15}$$

define a linear substitution of dimension n,

$$A = \begin{pmatrix} a_1^{(1)} & \cdots & a_n^{(1)} \\ \cdots & \cdots & \cdots \\ a_1^{(n)} & \cdots & a_n^{(n)} \end{pmatrix}, \qquad (x) = A(x'), \tag{16}$$

in which every row and every column contains exactly one non-zero coefficient, and that coefficient is 1. Hence $|A| = \pm 1$. If, after carrying out the substitution, one writes x_i again in place of x'_i, the result is just the permutation

$$\begin{pmatrix} 1, 2, \dots, n \\ \alpha_1, \alpha_2, \dots, \alpha_n \end{pmatrix}$$

of the indices.

Let B be another such substitution,

$$(x') = B(x''), \tag{17}$$

or, more explicitly,

$$x'_1 = x''_{\beta_1}, \quad x'_2 = x''_{\beta_2}, \quad \dots, \quad x'_n = x''_{\beta_n}. \tag{18}$$

Then composition gives

$$x_1 = x''_{\beta_{\alpha_1}}, \quad x_2 = x''_{\beta_{\alpha_2}}, \quad \dots, \quad x_n = x''_{\beta_{\alpha_n}}, \tag{19}$$

which is abbreviated by

$$(x) = AB(x''). \tag{20}$$

In other words,

$$\begin{pmatrix} 1, 2, \dots, n \\ \alpha_1, \alpha_2, \dots, \alpha_n \end{pmatrix} \begin{pmatrix} 1, 2, \dots, n \\ \beta_1, \beta_2, \dots, \beta_n \end{pmatrix} = \begin{pmatrix} 1, 2, \dots, n \\ \beta_{\alpha_1}, \beta_{\alpha_2}, \dots, \beta_{\alpha_n} \end{pmatrix}. \tag{21}$$

Thus, when the substitutions A and B are regarded as permutations of the indices, their product as substitutions agrees with their composite as permutations.

Therefore permutation groups on n letters are simply a special case of finite groups of linear substitutions of dimension n. Even permutations correspond to substitutions of determinant $+1$, and odd permutations to substitutions of determinant -1.

This connection between permutations and linear substitutions makes it possible to derive from every permutation group a group of linear substitutions, and conversely to treat certain questions about permutation groups by means of the theory of linear substitutions.